# Overview of Lab 3: Neural Networks and Model Evaluation

In Lab 3, we explored the implementation and evaluation of a machine learning model, focusing on the fundamentals of neural networks and the training process. The lab introduced how artificial neural networks can be used to learn patterns from data and make predictions for classification tasks.

The main concept covered was the structure of a feedforward neural network. We looked at how data passes through different layers of the network, including the input layer, hidden layers, and output layer. Each layer applies mathematical operations using weights and biases, with activation functions helping the model learn more complex relationships beyond simple linear patterns.

A major part of the lab involved understanding the training process of a neural network. We explored how the model makes predictions during the forward pass, calculates the error using a loss function, and updates its parameters through backpropagation and gradient descent. The learning rate was introduced as an important hyperparameter that controls how much the model adjusts its weights after each training step.

We also covered the importance of separating data into training and validation sets. The training set is used for learning the model parameters, while the validation set helps measure how well the model performs on unseen data during training. Concepts such as overfitting and underfitting were discussed, highlighting the need to balance model complexity and generalisation.

The lab also introduced practical PyTorch concepts, including creating models using `nn.Module`, switching between training and evaluation modes using `model.train()` and `model.eval()`, and using `torch.no_grad()` during validation and testing to prevent unnecessary gradient calculations.

Finally, we evaluated model performance using metrics such as accuracy and macro F1-score. These metrics provided a better understanding of how well the model performed, especially when dealing with multiple classes where accuracy alone may not fully represent performance.

Overall, Lab 3 provided practical experience in building, training, and evaluating neural network models. It connected the mathematical foundations of machine learning, such as optimisation and gradient descent, with the implementation process using modern deep learning frameworks like PyTorch.